

YANIS AKSAS

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EDUCATION

University College London (UCL)

Master Degree in Computational Finance

London, UK

Sep 2026 - Sep 2027

City University of Hong Kong

Exchange Program

Hong Kong, HK

Sep 2025 - Dec 2025

- Relevant coursework: Mathematic in Finance, Derivates & Risk Management, Data Engineering

Polytechnique Montreal

Bachelor Degree in Mechanical Engineering - Dean's list

Montreal, QC

Sep 2020 - Dec 2025

- Relevant coursework: Probability & Statistics, Differential equations, Numerical modeling

WORK EXPERIENCE

CITY UNIVERSITY OF HONG KONG

Research Assistant

Hong Kong, HK

Oct 2025 - Dec 2025

- Conducted literature review across multi-agent LLM trading systems and hyperparameter optimization methods to justify a systematic trading-optimization proposal for a Hong Kong brokerage firm, then co-designed the resulting 4-module workflow architecture (Strategist, Risk Manager, Memory Curator, Analyst) and multi-objective scoring framework

MMPROD

Software Developer

Montreal, QC

Sep 2024 - Feb 2025

- Built an offline desktop translation app for MMPROD using MarianMT, engineering a PDF and DOCX pipeline that preserved layout, tables, and images across 200+ page documents

FENIX

Software Engineer Intern

Montpellier, France

May 2024 - Aug 2024

- Designed and executed web scraping pipeline using Python and pandas to collect 9,000+ prospect contacts across France with geographic segmentation, building structured Excel Dataset for targeted marketing campaigns
- Redesigned company website using JavaScript with UX/UI optimization and implemented analytics tracking system, measuring 45% traffic increase and 85% conversion improvement post-launch

LEADERSHIP EXPERIENCE

POLY FINANCES

Software Developer

Montreal, QC

Jan 2025 - May 2025

- Collaborated on investment strategies as part of a student-run investment fund, conducting technical analysis and market research, presenting investment recommendations to expert committees for portfolio management decisions
- Developed and maintained a portfolio visualization application and job board platform (FinXplore) for the school's investment fund managing \$100,000+ in equity assets, using Python and Angular

DATA RANK

CTO

Montreal, QC

Dec 2024 - Present

- Co-founded DataRank as CTO, an invitation-based platform that scores a company's data maturity through AI-assisted questionnaires and interviews and generates a PDF report with prioritized recommendations for internal, client, or pre-audit review
- Built the platform's AI evaluation pipeline on an Express/Supabase backend with Stripe billing, including GPT-4 agents for pattern detection, benchmark comparison, and risk assessment

PROJECT EXPERIENCE

Deep Order Flow Imbalance (Python)

Montreal, QC

May 2026 – Aug 2026

- Replicated Kolm et al. (2021)'s Deep Order Flow Imbalance (OFI) paper on high-frequency return forecasting, engineering OFI features from 171M+ Nasdaq order book events (5 stocks, 32 days)
- Built a Ridge, MLP, and LSTM model progression in PyTorch, with the LSTM beating the linear baseline on 11 of 15 stock-horizon combinations

Energy Production Optimization Model (Python)

Montreal, QC

Sep 2024 - May 2025

- Built the network optimization module for HarmoniQ, a multi-source grid platform for Hydro-Québec, using PyPSA for grid topology and AC/DC power flow analysis, with linear programming (HiGHS solver) dispatch minimizing cost and CO2 emissions across 500,000+ hourly data points
- Co-built an interactive dashboard visualizing real-time network status, scenario planning, and dispatch results across hydro, wind, solar, nuclear, and thermal sources, enabling data-driven decisions for grid operators

SKILLS & INTERESTS

Programming: Python (NumPy, Pandas, TensorFlow, PyTorch), C++, R, SQL, MATLAB, JavaScript, Claude Code

Certificates: Machine Learning (Stanford), Financial Engineering & Risk Management (Columbia)